

# Author — AI Engineer Technical Assessment

3-day assessment · Gateway Digital

## The challenge

Given a user brief (topic, reader profile, length, tonality, genre), build an agentic system that produces a **publication-ready book** — with full front and back matter, multi-tonality voice, memory across chapters, and prose a human would actually want to read. We are not hiring for a pretty demo. We are hiring for how you think about **orchestration, grounding, memory, and voice**.

## MUST-have requirements

1. **Agentic pipeline, not one prompt.** At minimum: Planner, Researcher, Writer, Humanizer, Editor, Fact-Checker, Memory Keeper, Assembler. Document orchestration pattern (DAG, orchestrator-worker, event-driven) and coordination contracts.
2. **Publication-ready book.** Front matter: half-title, title, copyright (ISBN placeholder, edition, rights, CIP block), dedication, epigraph, TOC, foreword, preface, acknowledgments, introduction. Body: chapters with consistent structure and real callbacks. Back matter: afterword, appendix, glossary (from in-book terms), references, about-the-author, back-cover copy. Deliver PDF + DOCX with working TOC, roman-numeral front matter, page numbers from the introduction.
3. **Tonality as a first-class parameter.** Minimum 5 presets — Conversational, Academic, Storyteller, Motivational, Witty — cascading through every surface (prose, preface, afterword, glossary defs, about-the-author, back-cover copy), not only chapter bodies.
4. **Humanize — the book talks to the reader.** Second-person where tonality supports it, tonality-appropriate hooks, domain-drawn metaphors, varied sentence rhythm, memory-backed callbacks. Eliminate AI tells (“it’s important to note,” “delve into,” “in today’s fast-paced world,” “landscape of,” mechanical triads, symmetric “not only...but also”). Humanizer’s rules **MUST** appear in your prompt dossier.
5. **Memory across chapters.** Fact registry, concept/character bible, callback index, tonality fingerprint, decision log. A single context window does not count. Inserting a new chapter **MUST** trigger correct downstream repair (TOC, callbacks, glossary).
6. **Observability.** Per run: agent trace, complete prompt log, memory I/O log, token and cost ledger.

## AI capabilities we will evaluate

We want applied fluency, not acronym-dropping. Use what fits; justify each choice in your design log.

- **RAG.** Grounded facts for non-fiction chapters — chunking, embedding model, vector store, top-k, hybrid/dense/BM25 mix, reranking. No ungrounded freestyling on factual claims.
- **Tool use / function calling.** Agents calling retrieval, memory read/write, web search, fact-check lookups, structured extractors — with clean tool contracts.
- **Structured outputs.** Outlines, fact registry, callback index enforced via JSON schema or typed outputs, not regex over prose.
- **Preference-based refinement (RLHF / DPO / reward modelling).** You will not train a base model in 7 days; show applied awareness — preference pairs for Humanizer openings, a reward-model-lite scorer for tonality fidelity, or an LLM-as-judge loop with rubric prompts. We will ask how you would graduate this to full DPO/RLHF at scale.
- **Evals.** Automated checks for structural completeness, tonality fidelity (embedding distance to exemplars or LLM-judge), AI-tell detection, fact coverage, callback recall. Ship the rubric, runs, and scores.
- **Guardrails & hallucination control.** Fact-checker with abstention, citation-or-soften rule, no fabricated references implying real works.
- **Model strategy.** Mixed-model encouraged (larger for generation, cheaper for extraction and judging). Document the routing and cost rationale.
- **Context-window management.** Compression, summarization, memory-backed retrieval in place of stuffed prompts.

## Mandatory deliverables

- **Prompts dossier — weighted highest.** Every prompt for every agent, with purpose, inputs, outputs, known failure modes, and why this prompt. Hidden or reconstructed prompts = rejection.
- **Working MVP** — one-command runnable or hosted, with a sample brief ready to go.
- **Source repo** — honest commit history.
- **Architecture doc** — one diagram + one page: agent topology, memory stores, data flow, failure paths.
- **Memory schema** — concrete data models with example records.
- **Evals report** — rubric, runs, scores, failure analysis.
- **Trace bundle** — agent trace + prompt log + memory I/O + token/cost ledger for every test below.
- **Sample books** — PDF + DOCX for all four test cases.
- **Demo video** — 5–8 min, no edits that hide state.
- **Design decisions log** — ten most consequential calls and why.

### Test cases

- **A.** 10-chapter beginner's guide to personal finance, Conversational tone, ~2,500 words/chapter.
- **B.** 5-chapter novella, Storyteller tone, two named characters carried throughout.
- **C.** Take Test A's outline, regenerate Chapter 3 in Academic, Motivational, and Witty.
- **D.** Insert a new chapter between Test A's Chapter 4 and 5 — TOC, callbacks, and glossary **MUST** self-heal.

### Evaluation rubric

| Dimension               | Weight | 5/5 looks like                                                                  |
|-------------------------|--------|---------------------------------------------------------------------------------|
| Prompt engineering      | 25%    | Modular, role-clear, anti-vague, failure-mode-aware, evidence of iteration.     |
| Agentic design          | 25%    | Distinct agents, clean contracts, explicit orchestration, recoverable failures. |
| Memory management       | 20%    | Concrete schemas, correct I/O patterns per agent, passes insertion test.        |
| Book quality + humanize | 20%    | Reads as a real book; tonalities are genuinely distinct; AI tells are gone.     |
| AI capabilities         | 10%    | RAG, evals, guardrails, model strategy — applied and justified.                 |

### Rejection gates

Missing prompts dossier · single-prompt pipeline pretending to be agents · missing **MUST**-have book structure · tonality only affecting chapter bodies · no traces or logs · fabricated references implying real works · Test D continuity fails to repair.

### Submission

3 calendar days. Email: repo link, demo link, sample-books ZIP, and the prompts dossier as a standalone PDF.